5 Action-Perception as a Pattern Formation Process

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ABSTRACT

In this chapter we aim to identify collective variables (or order parameters) and their dynamics (stability, loss of stability, hysteresis . . .) for perception–action patterns. To do so we extend earlier results on phase transitions in human bimanual coordination to a perception–action task, synchronizing or syncopating finger flexion with an auditory metronome whose frequency is varied in steps (.25 Hz) over a wide range (1.0 Hz to 3.5 Hz). The chief idea is to study the metronome–hand relation as an example of a pattern forming system. *Relative phase* between hand and metronome proves to be an adequate collective variable that characterizes empirically observed patterns. A theoretical model using the concepts of synergetics and dynamic pattern theory reproduces all observed patterns, among which are frequency locking and phase locking, phase transitions, and loss of entrainment (relative coordination). The general implications of these results are discussed and new predictions proposed.

INTRODUCTION

Much has been written about the intimacy of the relation between perceiving and acting—that they evolved in parallel, and thus should be treated as a unitary system, and so forth (see e.g. Arbib, 1987; Gibson, 1979; Kelso & Kay, 1987; Kugler, Kelso, & Turvey, 1982; Lee & Young, 1986; Michaels & Carello, 1981; Warren, 1988 for recent treatments). Our understanding of this relation nevertheless remains quite limited. Although general in-

tuitions about the coherence of action and perception are useful, it is probably more important to find a specific case where these intuitions can be given a concrete and precise meaning. From that case, it may then be possible to generalize to broader contexts. The present research presents a detailed experimental and theoretical analysis of a specific, but nevertheless potentially generalizable situation—synchronizing one's motions with a regularly occurring environmental event.

Very many skills are of this type: from skipping, where one must synchronize one's action with the periodicity of the rope, to dribbling a basketball, to dancing, to stepping onto an escalator, and so on. In biology, numerous examples exist of external periodic stimuli entraining biological pacemakers—from firefly entrainment (Hanson, 1978) to regularly spaced synaptic inputs on "pacemaker" neurons (e.g., Perkel, Schulman, Bullock, Moore, & Segundo, 1964) to synchronizing central neurons in locust flight ganglia by peripheral sinusoidal driving of the wings (Wendler, 1974). We consider this problem as part of a broader research program dealing with dynamic pattern formation in complex, biological systems (Kelso, Schöner, Scholz, & Haken, 1987; Kelso & Schöner, 1987; Schöner & Kelso, 1988a,b,c). Our experimental results, nevertheless, may be understood in detail as corresponding to the process of entrainment of a nonlinear oscillator by an external harmonic source of excitation. Macroscopically, the patterns formed by the action–perception system correspond to empirically observed stable phasing (relative timing) relations and their breakdown. We believe these results to be quite relevant not only to a wider spectrum of skilled behavior (see General Discussion) but to a deeper understanding of information in the sense of information that is meaningful for biological functions and tasks (Schöner & Kelso, 1988c,e)—in short, semantic information (see also Shimizu, Yamaguchi, & Satoh, 1988).

What does it mean to understand action–perception as a pattern-formation process? From our perspective the key idea is to characterize action–perception patterns in terms of their dynamics. This means, fundamentally, identifying collective variables that correspond to these patterns and mapping them onto attractors of the collective variable dynamics. We can determine the dynamics of the collective variables by studying the stability and loss of stability of the patterns. Thus this dynamic pattern strategy (e.g., Kelso, et al., 1987; Kelso & Schöner, 1987; Schöner & Kelso, 1988a,b,c) can be implemented best near points of pattern change because these allow: (a) the clear differentiation of one pattern from another; and (b) the study of the relative stability of different patterns. More important, theoretically, is that such phase transitions enable us to formulate the (nontrivial) nonlinear collective dynamics underlying the patterns that may then be derived from a lower (individual component) level of observation. In fact, this strategy has been successfully applied (in the sense of an explicit

theory–experiment relation that can be formulated in closed mathematical form) to behavioral patterns that arise spontaneously without any specific environmental information (e.g., Haken, Kelso, & Bunz, 1985; Kelso, 1984; Schöner, Haken, & Kelso, 1986). More recently, the theory has been demonstrated to include certain environmentally elicited and learned patterns of movement coordination (Schöner & Kelso, 1988a,b) as well as intended behavioral patterns (see Kelso, Scholz, & Schöner, 1988; Schöner & Kelso, 1988c,e). The operational aspect of the theory cannot be overemphasized: In each case, dynamics are formulated for empirically observed variables and predictions made that can be experimentally checked. As a consequence of this work, a further, quite explicit meaning of the concept of information has materialized: Information is considered meaningful and specific only to the extent that it contributes to the collective variable dynamics, attracting the system to the required (or learned, or memorized, or intended) pattern (Schöner & Kelso, 1988c). Further clarification of this notion is provided in the theoretical modeling section of this paper, where we model explicitly the various perception–action behavioral patterns observed in the present study.

The present experimental configuration is somewhat similar to temporal tracking paradigms that have been studied spasmodically by psychologists and physiologists for over a hundred years. The task is to synchronize a movement, such as that of a finger, with an external stimulus such as a metronome or a strobe (see, e.g., Bartlett & Bartlett, 1959; Dunlap, 1910; Woodrow, 1932). Many of these older experiments were purely empirically motivated. For instance, Dunlap (1910) sought to determine the relationship between the length of an interstimulus interval in a metronome-following task and the directional error of response. More theoretically based efforts were conducted by Bartlett and Bartlett (1959) and more recently, (though in the same spirit of identifying central and peripheral processes involved in timing patterns) by Wing and Kristofferson (1973) and Hary and Moore (1987).

Based on previous theoretical and experimental work regarding the stability and change of coordinated rhythmic movements without explicit environmental information present, our interest was in the temporal order formed by the action–perception system. Can one determine the collective dynamics of this system also by studying the stability and (putative) loss of stability of the patterns formed? Might loss of stability be seen as a failure or entrainment limit, of synchronizing one's motion to the frequency of a periodically occurring event? And so on.

To investigate these kinds of questions, subjects were situated in a finger recording apparatus (see Kelso & Holt, 1980) and instructed to synchronize flexion movements (specifically, peak flexion) either "on the beat" or "off the beat" of an auditory metronome. The latter condition required subjects

to produce a flexion movement *between* (i.e., in the interstimulus interval) each metronome pulse whose frequency was varied either in an *ascending* order from 1.0 Hz to 3.5 Hz in steps of .25 Hz every 10 cycles or a corresponding *descending* order that began at 3.5 Hz.

The aim of the "off the beat" condition was to observe phase transitions—phenomenologically thought to be present[1]—in perception–action patterns, as a means of determining the underlying dynamics of these patterns. Thus, as mentioned briefly before, finding points at which the metronome-movement pattern changes not only allow one to distinguish one pattern from another, but the *relative stability* of the patterns can also be evaluated and quantified. Moreover, the mechanism thought to underlie changes in patterns is *loss of stability*, which can be measured given appropriate time scale relations (see Schöner, Haken, & Kelso, 1986, Kelso et al., 1987; Schöner & Kelso, 1988a,c). Other characteristic features of dynamic patterns, such as hysteresis, may be studied through comparisons of ascending and descending conditions. In the latter case, it also becomes apparent how the intentional production of behavior (Kelso, Scholz, & Schöner, 1988; Schöner & Kelso, 1988e) is constrained dynamically by the stability of the collective variable that characterizes the perception–action pattern.

METHODS

Subjects

Seven normal adults, five male and two female, volunteered for this experiment and were not paid for their participation. Five of the seven subjects were naive to the purpose of the experiment; two were not (SG and KS).

Apparatus

Flexion–extension movements of the subjects preferred index finger were recorded using an apparatus similar to the one described by Kelso and Holt (1980). The apparatus consisted of two manipulanda (one for each hand, although only one was used for each subject's preferred hand) connected to a metal sleeve into which the subject's finger was inserted. A linear potentiometer connected to the axis of rotation allowed for the direct transduction of the continuous angular position of the finger about the metacarpophalangeal joint as the finger cycled in the transverse plane.

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[1]The idea comes from a "test" that music teachers sometimes use. The student who maintains an "off the beat" pattern at the highest metronome frequency wins the game.

The stimulus, a periodic acoustic signal of adjustable frequency, was generated by a computer driven metronome constructed by the engineering staff at the Center For Complex Systems. The signal was output through an RS232 port and then conditioned to produce a train of 50 ms square wave pulses. This signal was in turn used to drive a loudspeaker producing a series of clicklike sounds. Both the square wave signal of the metronome and the finger position signal were recorded on FM tape and subsequently digitized at 250 Hz for later analysis.

Conditions

Subjects were instructed to synchronize peak flexion of the index finger (the point at which the movement of the finger changes from flexion to extension) with the metronome in either of two modes of coordination: (a) Off the beat (O)—so that peak flexion corresponds to the point in time midway between each metronome pulse, or (b) on the beat (I)—so that peak flexion occurred at the same time as each metronome pulse. The tasks were to be performed under two experimental conditions: (a) An ascending metronome series (A)—in which the frequency of the metronome was increased from 1 Hz to 3.5 Hz in the steps of 0.25 Hz, or (b) a descending metronome series (D)—where the metronome frequency was decreased from 3.5 Hz to 1 Hz in steps of 0.25 Hz. In all cases the metronome produced 10 cycles for each frequency plateau (e.g., at 1.25 Hz, 1.50 Hz, 1.75 Hz, . . .). Each mode of coordination was paired with each metronome series forming four conditions as follows: ascending–off the beat (AO), ascending–on the beat (AI), descending–off the beat (DO), and descending–on the beat (DI). Each subject performed a block of ten trials for each condition, and the order of conditions was randomized for all subjects.

Procedures

Subjects were seated comfortably in a chair in front of the apparatus. The height of the chair was adjusted to permit the forearms to rest horizontally on a padded support at the front of the apparatus. A soft adjustable plastic splint secured around the subject's forearm provided additional support and also served to restrain any unwanted movements of the wrist and digits. A soft thin pad was wrapped around the intermediate phalange of the index finger that was then inserted into the metal sleeve of the manipulandum up to the first interphalangeal joint. Correctly positioned, the sleeve prevented movements of the two distal finger joints. The position of the hand was then adjusted and fixed so that the center of rotation

of the metacarpophalangeal joint and that of the manipulandum were colinear.

After positioning the subject in the apparatus, instructions were issued. The subject was given a description of the conditions and instructed to cycle his/her finger once for each beat of the metronome maintaining the required mode of coordination as well as possible. Subjects were further instructed that should they feel the pattern begin to change they should not try to resist. It was emphasized that the primary task was to follow the metronome, maintaining one cycle of the finger for each cycle of the metronome. A demonstration of peak flexion on the beat and off the beat was given to insure that subjects clearly understood the task. A single trial run, 11 frequency plateaus, 10 cycles per plateau lasted approximately 1 min. Rest intervals were provided to the subject after each trial block of 10 runs, or on request. The time for the experiment varied between 1.5 and 2 hours.

Analysis

Following digitization of the data, special in-house wave analysis software was employed to obtain the mean frequency of metronome and finger movement, the mean relative phase between the finger and the metronome, and the standard deviation of these variables at each frequency plateau for each subject performing in each of the four experimental conditions.

These four variables were then treated as individual scores in later analyses. In order to eliminate transient effects resulting from the change in metronome frequency (i.e., while subjects adjusted to the change) events occurring within the first 2 cycles of the metronome pulse at each frequency plateau were excluded in the calculations. The frequency of finger movement was obtained by computing peak flexion to peak flexion–cycle durations. Relative phase calculations were obtained by computing the position in time of peak flexion of the finger with respect to the local peak-to-peak cycle of the metronome. Depending on the relative phase behavior at a given plateau, one of the following modes of analysis was used to calculate the mean and standard deviation: (a) the 180 degree mode calculates relative phase with respect to a 0 to 2π reference interval; (b) the zero degree mode calculates relative phase between $-\pi$ and $+\pi$; and (c) the absolute mode calculates relative phase in the interval $-\pi$ to π without regard to sign. If the relative phase at a given plateau was centered around 180 degrees, then the 180 degree mode was used. If relative phase centered around zero degrees, then the zero mode calculation was used. In cases where the relative phase was changing, the absolute mode was used.

Correlations between required (metronome) and actual (finger move-

ment) frequencies were calculated for each subject in all four experimental conditions. Analyses of variance (ANOVA) were conducted separately for the ascending and descending conditions (see Results). For each subject, a mode by plateau ANOVA was conducted for pre- and post-transition plateaus treating trial runs as cases. Because most subjects differed in the exact frequency at which the switch from one pattern to another occurred, the number of pre- and post-transition plateaus varied in this analysis. Also, because of within-subject variation in switching frequency across trials, a minimum of 5 trials was required at each frequency plateau in order for that plateau to be included as a variable in the ANOVA.

RESULTS AND DISCUSSION

Accuracy of tracking

First, we establish that subjects were capable of tracking the metronome across all the frequencies used (1.0 to 3.5 Hz) and in all conditions. Pearson's product–moment correlations between actually produced and required frequency were calculated for all subjects. The means for ascending (in phase and antiphase) conditions were $r = .98$ (SD = .024) and $r = .99$ (SD = .013). For the descending conditions, the respective correlation values were $r = .996$ (SD = .004) and $r = .98$ (SD = .015). The range of correlations across subjects and conditions was $r = .933$ to $r = 1.0$. The overall correlations, though strong, can be a bit misleading under more detailed analysis in that under certain circumstances the frequency of finger movement does not conform to the frequency of the metronome. Obviously, when a transient in relative phase occurs, the frequencies must differ. Also, a necessary condition for the absence of phase-locking, that is, when the relative phase "drifts" or "wanders," is frequency detuning. We consider these cases in more detail in the next section.

Patterns of Behavior: Ascending Conditions

Before providing quantitative results, we present a classification of the patterns produced by subjects when "prepared" initially in the in phase and antiphase modes of coordination. For the in phase synchronization condition this was quite easy. All 7 subjects on the majority of runs (68/70 or 96.6%) were capable of maintaining a synchronous, in phase relationship with the metronome across (at least) part of the range of frequencies. An example of the metronome and hand movement time series and the relative phase between them is shown in Fig. 5.1 (top). There it can be seen that finger flexion coincides (phase-locks) with the metronome

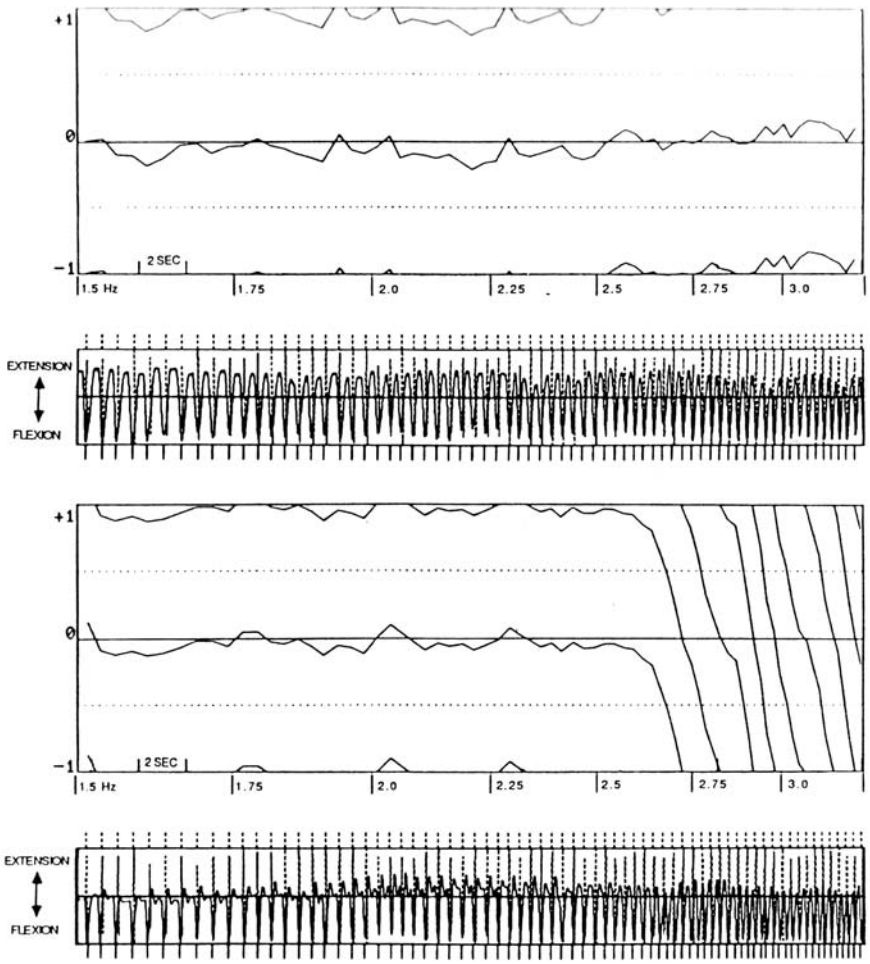


FIG. 5.1. Ascending in phase condition. Representative examples of metronome and hand movement time series and the relative phase between them are shown as a function of required frequency (see text for details). Subjects are: SG (top); BJ (bottom). Units of relative phase are radians divided by 2π.

as metronome frequency changes. On 14/70 trials, we also observed a loss of phase locking at higher frequencies; that is, subjects (primarily one, 7/10 trials) were unable to synchronize in the in-phase mode at higher frequencies and phase-drift was observed. Only a very small frequency difference (or detuning) between the metronome and finger is necessary to induce phase drift. An example of wandering at higher frequencies is presented in Fig. 5.1 (bottom).

For the ascending antiphase condition, a variety of patterns were observed. Only one subject (LS) was able to preserve an antiphase relation with the metronome across all frequency plateaus. It turned out however, on postexperiment interview, that this subject discovered a special way to perform the task. Thus, instead of performing a *flexion* movement *between* metronome pulses as instructed, the subject produced an *extension* movement in synchrony with the metronome pulse. Using this strategy, the subject had little problem producing a flexion movement off the beat, that is, an antiphase relation with the metronome. Thus, in all ten trials, this subject stayed in the antiphase pattern with the metronome.

Sidestepping these results for the moment (see General Discussion), let us consider other action–perception patterns in the ascending and antiphase condition. On 37% of the trials, a spontaneous transition from an antiphase to in phase pattern was observed (see Fig. 5.2). Five of the six subjects (excluding LS) exhibited this pattern on three or more trial runs (Fig. 5.2A). On an additional 33% of the trials, subjects—after switching to the in phase pattern and stabilizing there—proceeded to display phase-wrapping behavior, in which the relative timing between metronome and finger was not stationary. Four of the six subjects showed this additional feature, one subject on all ten runs (see Fig. 5.2B).

On a proportion of trials (18%), we observed a direct pathway from the initial antiphase pattern to phase wrapping, mostly in one subject (6 out of 10 trials, see Fig. 5.2C). On a further 11% of trials, after phase wrapping, the pattern shifted to in phase and stabilized there for the remainder of the trial run (see Fig. 5.2D). In sum, on a total of 80% of the trials, subjects showed a transition from antiphase to in phase entrainment. Phase drift was observed *en route* to the in phase pattern but mostly following the in phase pattern.

Patterns of Behavior: Descending Conditions

Once again, for the in-phase condition there was little problem in classifying patterns of relative phase between finger and metronome. On 65/70 trials (93%) subjects maintained an in phase relation at all frequencies. On 2/70 (3%) phase drift was observed before locking occurred. On a further 3 trials (2/10 for one subject and 1/10 for the other), the pattern went from in phase to antiphase and back again.

None of the subjects, save one, were able to maintain an antiphase relation with the metronome at higher frequencies. The "outlier" (LS), however, was able to do so because of adopting the strategy of producing extension on the beat, rather than flexion off the beat (see previous discussion). Even with this strategy, on 7/10 trials this subject displayed either phase drift before achieving the required phase relation, or adopted the

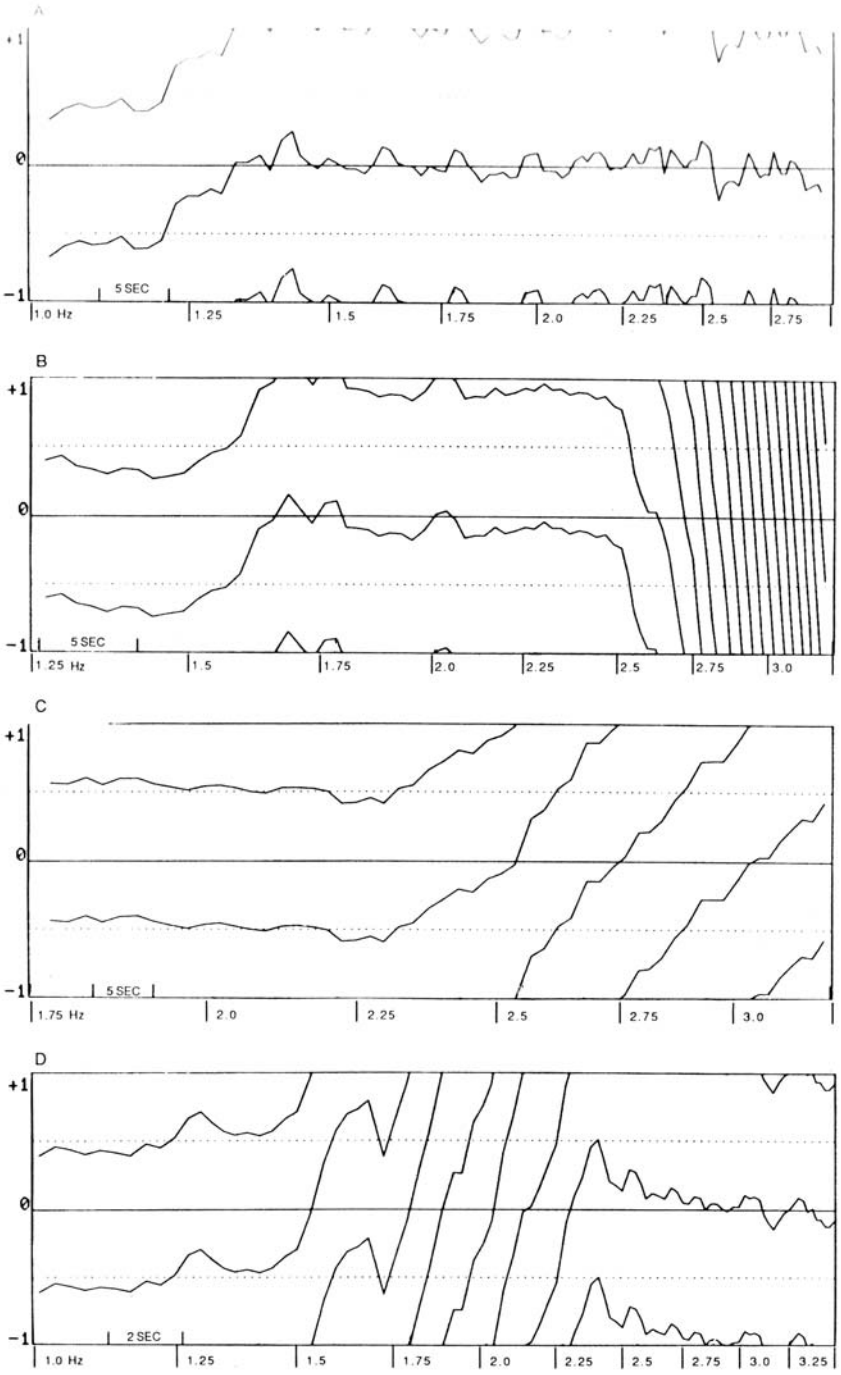


FIG. 5.2. Ascending antiphase condition. Representative examples of the relative phase between metronome and hand movement are shown as a function of required frequency for different subjects: (A) SG; (B) BJ; (C) JJ; (D) KS.

in-phase pattern first, that is, at higher frequencies, before shifting to antiphase.

All of the remaining subjects were able eventually to maintain an antiphase relation with the metronome. The pathways to this perception–action pattern varied, however. On 20% of trial runs, as frequency decreased, phase drift was followed by an in phase pattern followed by the antiphase pattern. In 8% of the cases, no phase drift was observed: The system changed from in phase to antiphase. A further 43% of trials showed a direct change from phase drifting to the antiphase pattern. Forty-eight percent of the latter runs were preceded by an in phase relation between movement and metronome. A few other patterns were observed, including ones in which several bouts of phase wandering occurred *en route* to the required pattern. Fig. 5.3 displays a number of examples of the descending patterns that are discussed more thoroughly in the Modelling section.

Quantitative Analysis: Ascending Conditions

We examine the pretransition régime first for effects of mode or initial condition (in phase or antiphase), frequency plateau, and their interaction, on the measures of relative phase between hand and metronome, cycle duration, and their respective variabilities. Only significant effects are mentioned. As expected, the mode effect on mean relative phase was significant for all subjects. Individual F-values ranged from 166.35 in BJ to 5516.42 in JJ ($ps < .01$). In general, the longer the subject maintains the less-stable antiphase relation, the greater the level of relative phase fluctuations. Two of the subjects' data (TB and MC) tend to support this contention: A mode by plateau interaction was observed, $F(4,82) = 5.45$, $p < .01$ (TB) and $F(6,115) = 2.47$, $p < .03$ (MC) respectively. This interaction for TB (A) and MC (B) is shown in Fig. 5.4. For TB, the difference between modes is statistically significant on plateau 3 (1.5 Hz) and plateau 5, the pretransition plateau (2 Hz). For MC, the antiphase mode is more variable for plateaus 5 (2 Hz) and the pretransition plateau (2.5 Hz). None of the other subjects, however, showed a mode or mode by plateau interaction on relative phase variability. This is not surprising because subjects KS and BJ already revealed transition behavior on the third frequency plateau. In subject SG, nearly all transitions occurred on the second frequency plateau, making a detailed pretransition analysis impossible.

Subject LS who displayed no transitionlike behavior because of the strategy adopted, required a separate statistical treatment. Effects of mode, frequency plateau, and the interaction between these variables were found for mean relative phase, $F(1,194) = 2952.28$, $F(10,194) = 3.26$ and $F(10,194) = 3.44$, $p_s < .01$ respectively. The interaction was largely due to the strong main effects found for both frequency and mode (see Fig. 5.4, bottom).

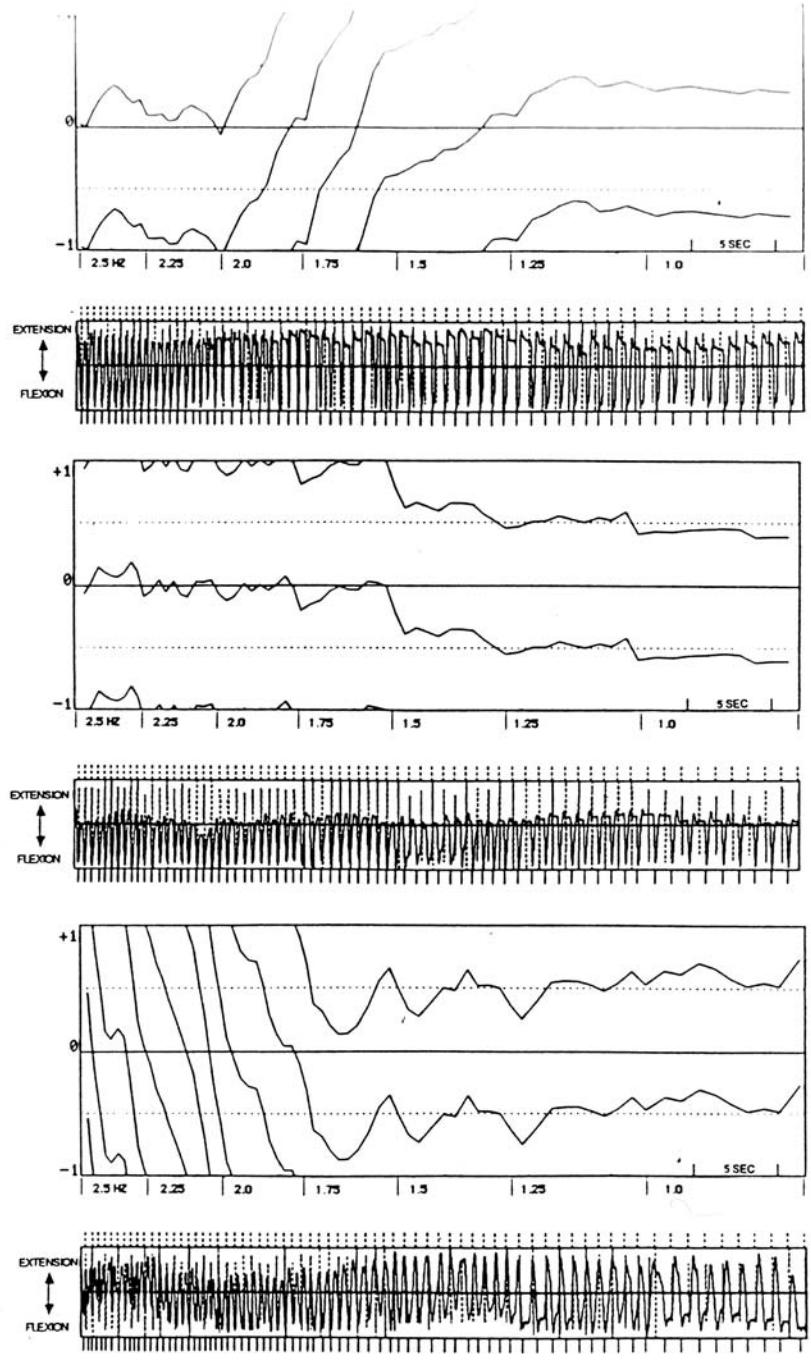


FIG. 5.3. Descending antiphase condition. Representative patterns as frequency decreases are shown for different subjects: SG (top), BJ (middle); TB (bottom) (see text for details).

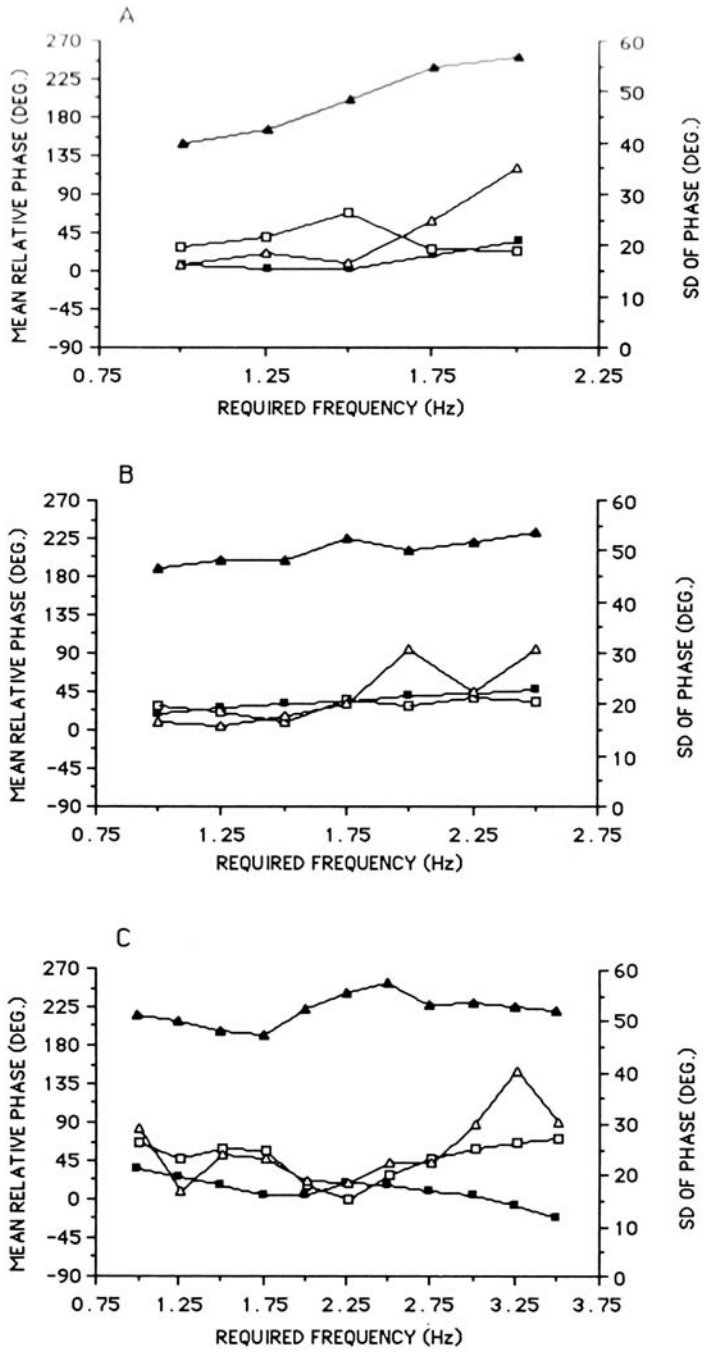


FIG. 5.4. Ascending antiphase (triangles) and corresponding in phase (squares) conditions: The mean relative phase (filled symbols) and the standard deviation of relative phase (open symbols) are shown as a function of the required frequency for the pretransition frequency plateaus. Subjects are: TB (a); MC (b); LS (c).

Phase variability increased with frequency, $F(10,194) = 6.05$, $p < .01$, but was not differential between modes. Typical effects on cycle duration and its variability with frequency plateau $Fs(10,194) = 12087.54$ and 31.94, $p_s < .01$ were found, as well as a mode by plateau interaction, $F(10,194) = 2.30$ and 2.03, $p < .05$.

For the 6 other subjects, modest influences of frequency on mean relative phase were observed as well as an interaction with mode. In 3 subjects (TB, BJ and JJ) the interaction was significant, $F(4, 82) = 8.30$, $F(1.32) = 10.05$, and $F(6,114) = 32.00$, $p_s < .01$ respectively. A main effect of frequency plateau was also noted for MC, $F(6,115) = 8.78$, $p < 01$. Although we expect the mean phase to be different between in-phase and anti-phase modes at all frequency plateaus, differential effects of frequency between the two modal conditions were not expected. *Frequency* as used here is meant to be nonspecific, in the sense of simply moving the metronome-hand system through different collective states or patterns. The reasons for systematic effects of frequency on relative phase need to be studied further, though we suspect they are largely artifactual and due, in part, to the classification scheme we employed: phase means designated to be equivalent across frequency may not have been equivalent after all.

It was no surprise that the frequency plateau influenced cycle duration in all subjects ($p_s < .01$). An interaction of mode and plateau was noted for two subjects (TB and MC) and a mode main effect on one (KS), $F(4, 82) = 3.31$, $p < .02$, $F(6, 115) = 2.26$, $p < .05$ and $F(1, 33) = 4.22$, $p < .05$ respectively. Overall, these effects are consistent with the phase results: Small differences in cycle duration between modes are indicative of frequency detuning between metronome (whose frequency is constant on a plateau) and hand, giving rise to phase variation. As cycle duration decreased with required frequency, its variability also decreased—a well known result (e.g., Deutsch, 1983), $F(1, 33) = 12.05$, $p < .01$ (KS); $F(6, 114) = 10.97$, $p < .01$ (JJ); $F(1, 35) = 5.52$ $p < .03$ (BJ); $F(6, 115) = 18.77$, $p < .01$ (MC); $F(4, 82) = 9.41$, $p < .01$ (TB). In some subjects, mode aslo had an effect on cycle duration variability, $F(1, 33) = 4.28$, $p < .05$ (KS); $F(1, 114) = 4.82$, $p < .03$ (JJ); and $F(1, 35) = 5.52$, $p < .03$ (BJ); and an interaction between mode and frequency plateau was found for another (MC), $F(6, 115) = 2.39$, $p < .03$. Variability in cycle duration tended to be greater in the antiphase condition than in the in phase condition, and was enhanced to a greater extent before the transition in subject MC.

To the extent subjects' behavior stabilized after changing pattern, we were able to perform a posttransition analysis analogous to the one described above. Small mode differences in mean phase and phase variability were noted in 2 subjects, both of whom exhibited a transition to the in

phase pattern at a low frequency, $F(1, 55) = 10.95$ and 9.60, $p < .01$ (BJ) and $F(1, 114) = 47.17$ and 10.60 (SG) respectively. Still another subject (KS) exhibited a mode by frequency plateau interaction on both the mean and variability of relative timing, $F(5, 105) = 3.95$, $p < .01$ and 2.83, $p < .02$ respectively. Again, these results are linked to changes in cycle duration and its variability after the transition is over. The same subjects revealed a mode effect on cycle duration, $F(1, 55) = 10.67$, $p < .01$ (BJ) and $F(1, 114) = 12.86$, $p < .01$ (SG). In addition, the variability of cycle duration increased as frequency increased in posttransition regimes, $F(5, 105) = 5.18$ (KS), $F(6, 114) = 10.97$ (JJ), $F(3, 55) = 4.90$ (BJ), $F(6, 114) = 9.73$ (SG), $F(6, 115) = 18.77$ (MC) and $F(4, 82) = 9.41$, $p_s < .01$ respectively. The modes were a little different in cycle variability in one subject (JJ), $F(1, 114) = 4.82$, $p < .03$, after the transition even though they are supposed to be the same. One might interpret these differences as "history" dependent effects: Where you come from has an influence on the stability of the performed pattern.

Quantitative Analysis: Descending conditions

As one might expect from the foregoing qualitative description of the descending patterns, it does not make much sense to try to perform the same kind of pre- versus posttransition analysis as we did for the ascending conditions. Nevertheless, it is possible to characterize the patterns quantitatively at lower frequencies, and there are a number of new aspects in the evolutionary path toward the timing patterns produced at lower frequencies.

It is important to recognize that the two cases—ascending and descending—are very different. In the former, it is possible to prepare the system in an antiphase pattern, for example, and watch it evolve as frequency increases. In the latter, as frequency decreases, the subject must "probe" for the antiphase pattern and once he/she finds it, the task is to maintain that pattern. There are thus many more transients and phase-wrapping phenomena in the antiphase descending condition than in any other condition. To perform this task, the subject must first track the metronome, that is, frequency lock, and then phase lock in an antiphase relation. Of course, as described earlier, some subjects do both in parallel, that is, they frequency- and phase-lock simultaneously, often after a bout of phase wandering or drifting. Fig. 5.5 shows an example of the phasing patterns between hand and metronome in the descending anti-phase condition, as they evolve over time (trials). This subject (JJ) had little or no problem synchronizing in phase "on the beat," in the descending condition, as can be seen in his very first trial run (see Fig. 5.5A)

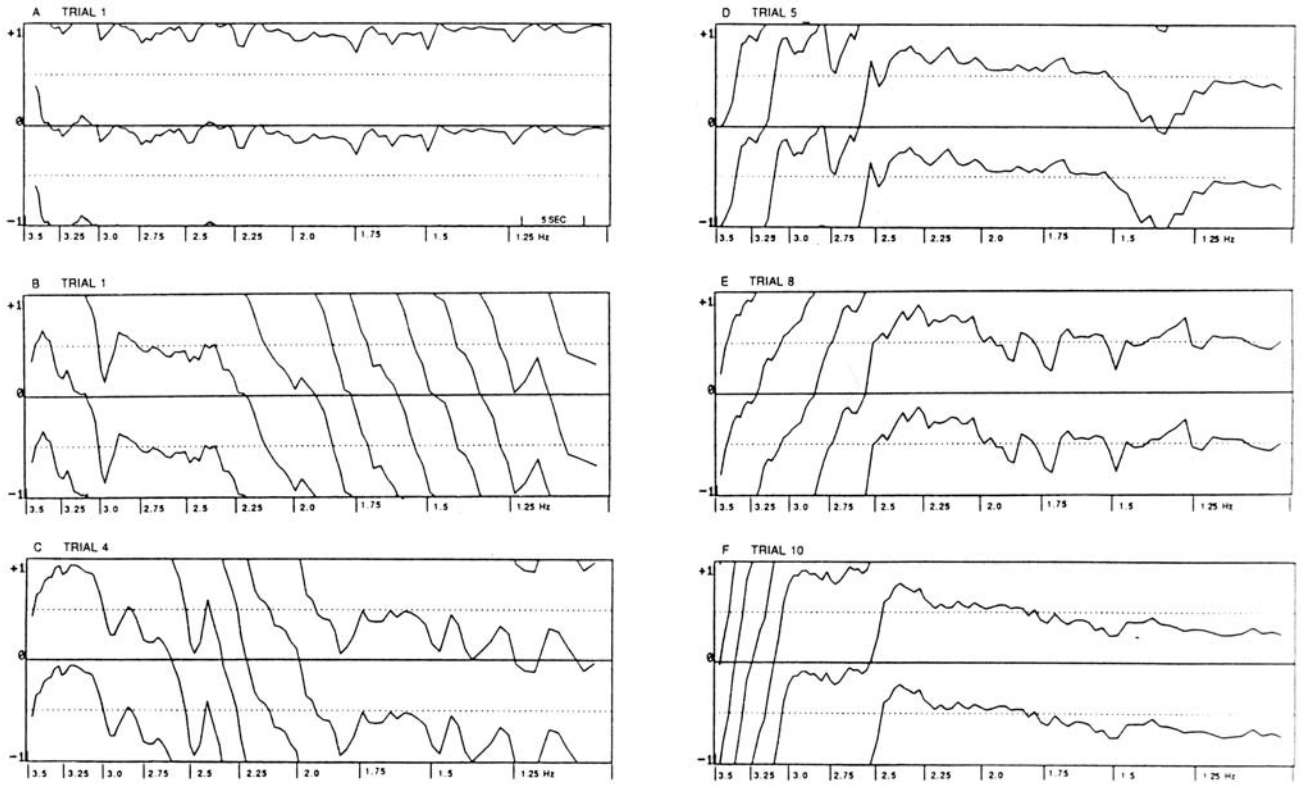


FIG. 5.5 Descending antiphase condition (Panels B through F). Time series of relative phase illustrate the patterns of behavior over subsequent trials of subject JJ. The decreasing frequency within each trial is indicated. Note the earlier onset of the required antiphase patterns for later trials. In Panel A a descending in phase trial is shown for comparison.

It is clear also from Fig. 5.5 that early trials are dominated by phase-drift, but that the extent of such drift diminishes over time (trials). Correspondingly, the amount of time spent on producing the required anti-phase relation increases systematically. The degree of deviation or fluctuation around the antiphase state also decreases (see posttransition analysis below). The attraction of the in phase pattern is apparent throughout, but its influence also diminishes—even over the space of a single trial (compare e.g., trials 4 and 5 of Fig. 5.5). Very similar behavior is observed in other subjects, though perhaps performance improves more gradually than in subject JJ. The histogram shown in Figure 5.6A shows a steady increase across trials in the number of antiphase patterns produced, as well as their progressively earlier appearance in time (TB).

Still other subjects appear to be entirely frequency dependent, that is, it is only when a certain cycling frequency is reached that the antiphase pattern appears. An example is shown in Fig. 5.7 (KS). Phase wrapping is present (the slope of which reflects the frequency difference between hand and metronome) until, essentially, the penultimate frequency plateau is reached, when the antiphase pattern is observed. A very similar result is shown for two other subjects (BJ and SG) in histogram form in Fig. 5.6 (B and C). Note that regardless of the number of trials, these subjects appear to reach a certain frequency plateau before the antiphase relation is stably produced. Thus there is little or no "learning" or "adaptation"—in the sense of ability to perform the antiphase pattern at an earlier, higher frequency—as was observed for JJ and TB (Fig. 5.5 and 5.6A).

As one might expect, those subjects who changed to the antiphase pattern exhibited a strong mode effect on mean relative phase. For F-values ranged from 5819.54 (MC) to 314.07 (BJ), $p_s < .01$. In addition, phase variability is enhanced in the antiphase, relative to the in phase pattern in 3 of the subjects (including LS) as revealed by a mode by frequency plateau interaction, $F(7, 137) = 2.24$, $p < .03$ (MC), $F(3, 54) = 4.94$, $p < .01$ (JJ) and $F(9, 166) = 1.96$, $p < .05$ (LS). These interactions are plotted in Fig. 5.8 for all 3 subjects. For MC, the antiphase pattern is statistically more variable than the in phase pattern on the first three posttransition plateaus (Fig. 5.8A); for JJ, only the first posttransition plateau is statistically different (Fig. 5.8B); and for SL (Fig. 5.8C), although the variability of the antiphase pattern is greater in the first four posttransition plateaus, only on the first one is the variability statistically different between the two patterns.

Analogous to ascending conditions, as cycling frequency decreased over time in the posttransition régmie, its variability increased for all subjects, but one. Also, subject LS showed an interaction between plateau and mode, $F(9, 166) = 1.68$, $p < .01$) but the difference between modes reached significance only at the slowest (1 Hz.) frequency.

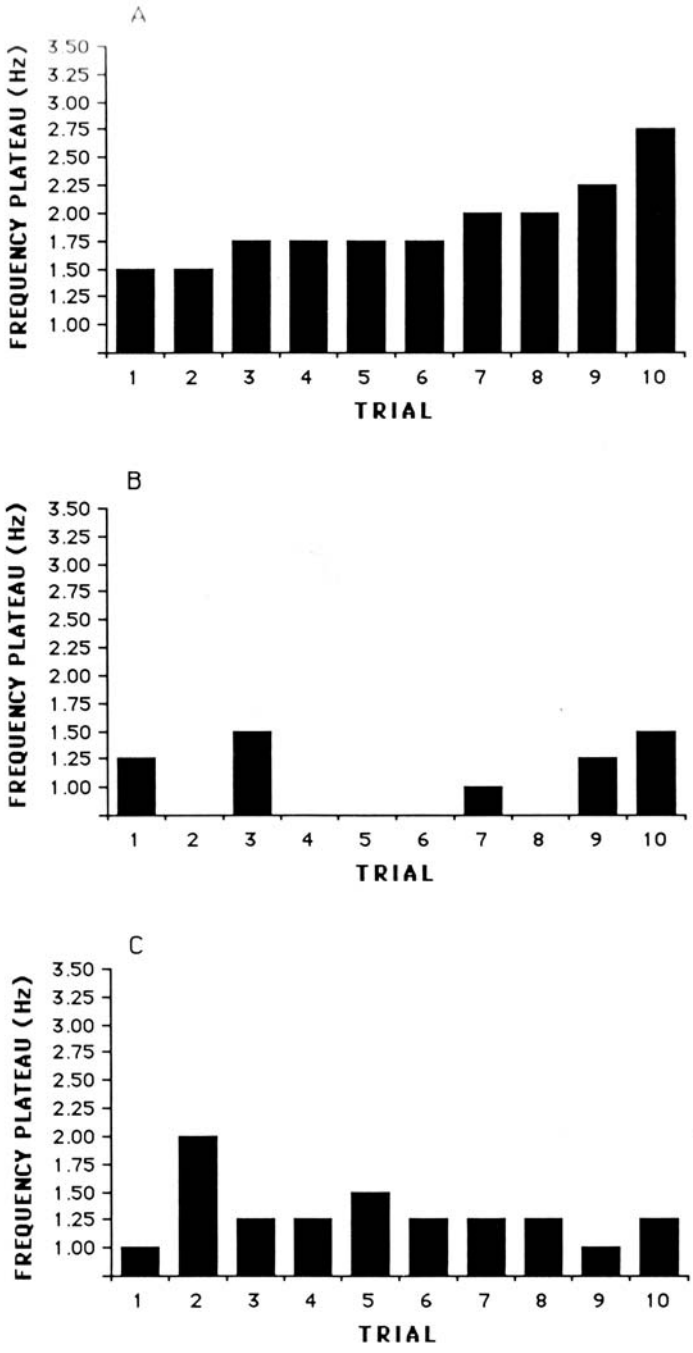


FIG. 5.6. Descending antiphase condition: The darkened bars designate those frequency plateaus at which subjects were able to maintain the antiphase pattern. Note that as a function of trial subject TB (panel A) reaches the antiphase pattern at progressively higher frequencies, whereas subjects SG (panel B) and BJ (panel C) show no such progression.

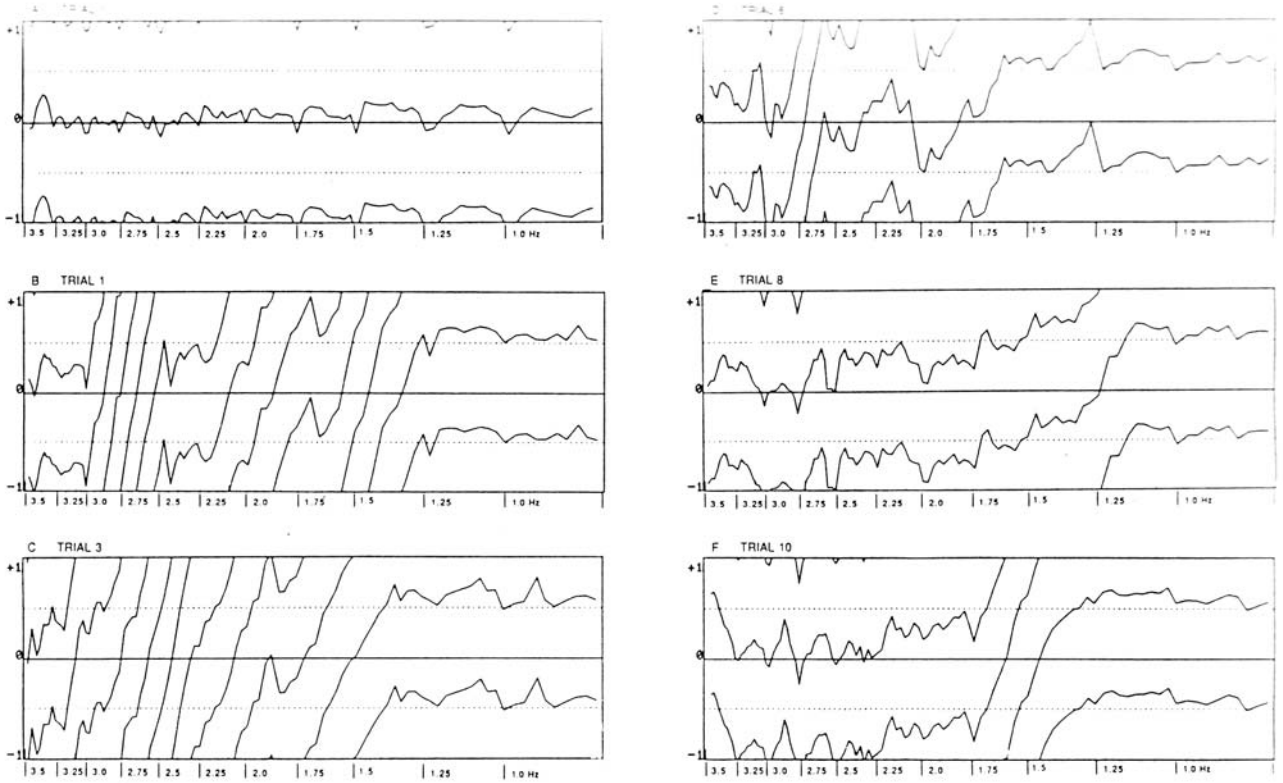


FIG. 5.7. Descending antiphase condition (Panels B through F). Time series of relative phase illustrate the patterns of behavior over subsequent trials of subject KS. The decreasing frequency within each trial is indicated. Note that the onset of the antiphase pattern occurs at a similar frequency across trial runs. In Panel A a descending in phase trial is shown for comparison.

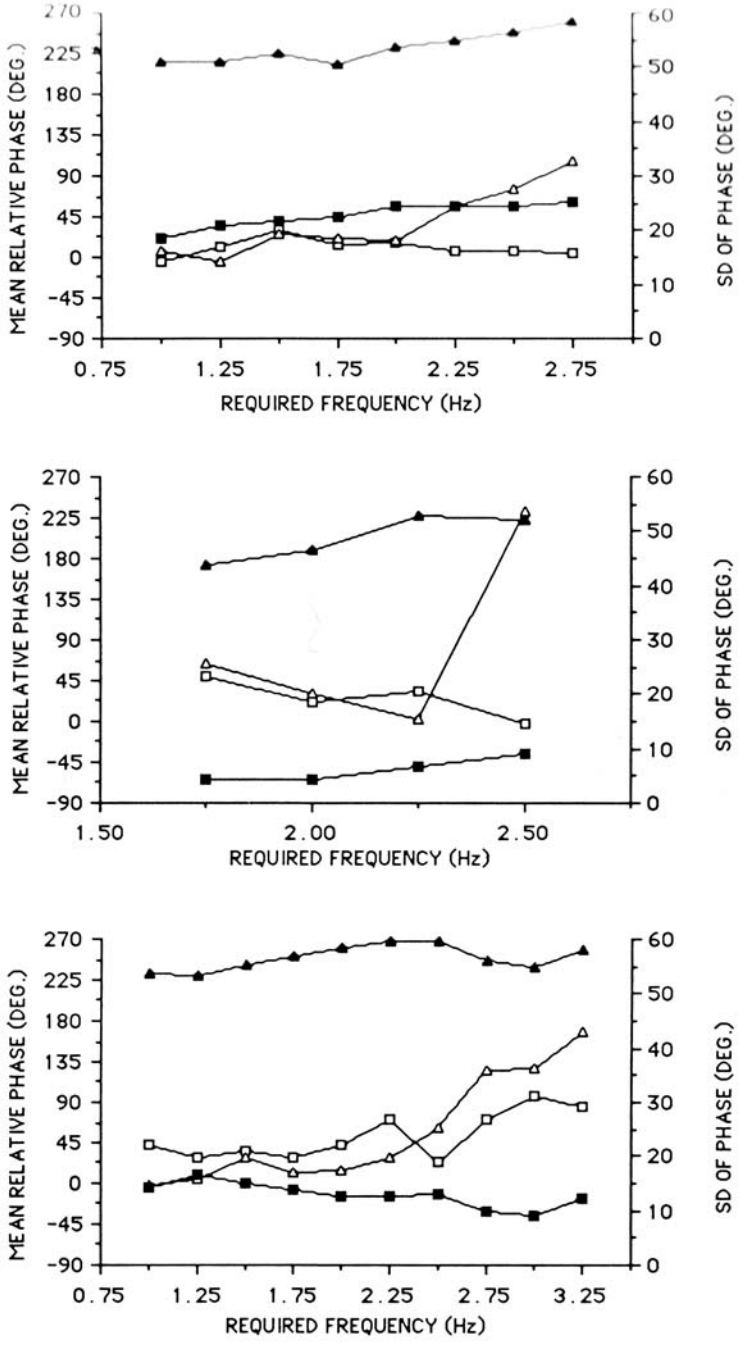


FIG. 5.8. Mean relative phase (filled symbols) and standard deviation of relative phase (open symbols) for posttransition frequency plateaus in the descending antiphase (triangles) and in phase (squares) conditions. Subjects are MC (top), JJ (middle) and LS (bottom).

Hysteresis

It is intriguing to inquire, even though the tasks for ascending and descending conditions are very different, whether subjects switch patterns at the same frequency. The frequency plateau at which the subject switched permanently from one pattern to the other was noted and *t*-test comparisions conducted on each subject's data. The results were not very systematic: On the average 1 subject switched at a lower frequency plateau in the ascending condition than the descending condition, $t(12) = -3.52$, $p < .004$. The opposite occurred for another subject, $t(14) = 2.22$, $p < .04$. The remaining subjects showed no reliable difference between ascending and descending conditions.

A THEORETICAL MODEL

To determine a theoretical model of the present action–perception system within the framework of the dynamic theory of behavioral patterns (c.f., Kelso & Schöner, 1987; Schöner & Kelso, 1988c) we identify in a first step an *order parameter* in the sense of synergetics (Haken, 1983) and its dynamics, thereby choosing a level of observation. As discussed earlier, the relative phase, ϕ, between metronome and rhythmically moving limb is a good candidate. Because both metronome and moving limb are periodic in time, this relative phase is theoretically well defined.

An illuminating subtlety, not addressed so far, involves the exact definition of relative phase. Such definition determines, for example, which pattern is called in phase and which pattern antiphase. Is this choice purely conventional? The answer lies in the functional specificity of the order parameter dynamics. The task requires synchronization of a specific aspect of the movement—here, peak flexion—with the metronome signal. We define the relative phase measure such that temporal alignment of the metronome pulse with this specific movement event corresponds to an in phase pattern. Thus, in the present experiment, peak flexion is used as the reference event from which relative phase is calculated (see Methods). As a result, "on-the-beat" flexion corresponds roughly to in phase and "off-the-beat" flexion corresponds roughly to antiphase. We assume that the dynamics are specific to the task and independent of the synchronization strategy, whereas the *meaning* of relative phase depends on this strategy. If, for example, subjects were instructed to synchronize extension with the metronome rather than flexion, we would expect that the transition phenomenon would be inverted: Events that previously played the role of in phase now become antiphase and hence correspond to the less stable pattern. Whether this view is correct can obviously be tested experimentally.

Because oscillatory states for the collective variable, relative phase, appear not to be relevant, we assume that the dynamics of relative phase are of first order. To find a simple dynamical system that takes the angular character of relative phase into account, we expand the equation of motion into a Fourier series, keeping only the first few terms. It is important to note that we cannot assume symmetry of these dynamics under the operation $\phi \rightarrow -\phi$, because metronome and moving limb are, of course, functionally different. Limiting the expansion to second order we find:

$$\dot{\phi} = \Delta\omega - a \sin(\phi - \psi_1) - 2b \sin(2\phi - \psi_2) + \sqrt{Q}\xi_t \qquad (1)$$

where $\Delta\omega$, a, b, ψ_1 and ψ_2 are constants. For reasons discussed elsewhere (e.g., Schöner, Haken, & Kelso, 1986) we have added a Gaussian white noise process, ξ_t, as a stochatic force of strength Q.

A theoretical description of this perception–action system can also be obtained on the level of the component oscillator. On that level of description, the finger position, x, and its velocity, $\dot{x}$, play the role of order parameters. The observed rhythmic behavioral pattern is mapped onto a limit cycle of the $(x, \dot{x})$ dynamics (Haken, Kelso, & Bunz, 1985; Kay, Kelso, Saltzman, & Schöner, 1987). Using the concept of environmental information (Schöner & Kelso, 1988a,c) the influence of the metronome signal can be captured as a contribution to the oscillatory dynamics (see Schöner & Kelso, 1988b, for a related theoretical analysis). From such an underlying oscillator model we find that the parameter $\Delta\omega$ is due to the difference between the metronome frequency and the frequency that the moving limb generates spontaneously. As this frequency difference widens, synchronization is lost and running solutions are found. Futhermore, deriving the relative phase dynamics, Equation (1), from the oscillator model, we find $\psi_1 = \psi_2 = 0$. Therefore we neglect ψ_1 and ψ_2 in the following.

It is fairly easy to obtain an intuition about the solutions of Equation (1). In Fig. 5.9 we have plotted the right hand side of this equation of motion as a function of $\dot{\phi}$ for various parameter values. The system has stationary states where $\dot{\phi}$ crosses the ϕ-axis. These are stable states if the slope of $\dot{\phi}$ is negative, and are unstable states if the slope is positive at this point. The arrows in Fig. 5.9 indicate the direction of the flow. For $\Delta\omega << a$ and $\Delta\omega << b$ (Fig. 5.9, left) two stable states exist, one close to in phase and one close to antiphase. If, for example, b is decreased, the amount of attraction to these stable states decrease until one (Fig. 5.9, center) and ultimately both (Fig. 5.9, right) cease to be stationary solutions. When stationary solutions are no longer found, "running" solutions (ϕ ever increasing or decreasing) or, if one applies the periodicity convention to relative phase, "wrapping" solutions occur. Note that for $\Delta\omega = 0$ the system is identical to the relative phase dynamics found earlier for interlimb coordination patterns (Haken, Kelso, & Bunz, 1985). In that case,

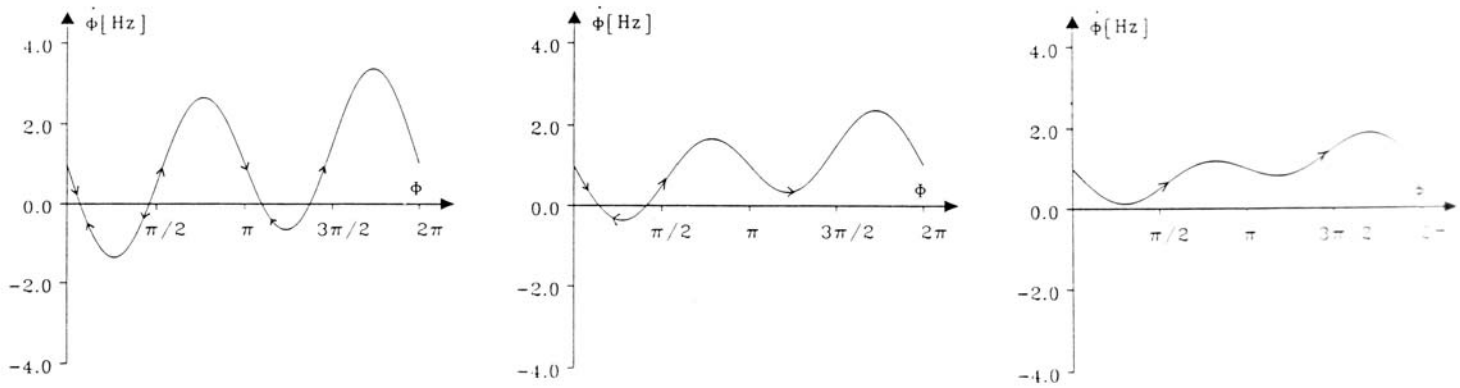


FIG. 5.9. The right hand side of the equation of motion Equation (1) plotted as a function of relative phase for a = 0.5 Hz., $\Delta\omega$ = 1 Hz and b = 1. Hz (left), b = .5 Hz (center) and b = .25 Hz (right). The arrows indicate the flow of $\dot{\phi}$. Zeros of $\dot{\phi}$ are stationary solutions that are stable if the slope is negative and unstable if the slope is positive at the stationary state.

the system has either one attractor at $\phi = 0$ or two attractors at $\phi = 0$ and $\phi = \pm\pi$.

To obtain a survey over the possible states and their stability, we show in Fig. 5.10 a phase diagram of Equation (1), that is, those regions in parameter space in which the same number of states are stable as well as the boundary lines between different such regions. The stationary states, ϕ_0, were determined by solving numerically

$$\dot{\phi} = \Delta\omega - a \sin(\phi_0) - 2b \sin(2\phi_0) = 0 \tag{2}$$

Only attractors, that is, stationary states with positive relaxation rate, $\tau_{rel}^{-1} = a \cos(\phi_0) + 4b \cos(2\phi_0)$, are considered. In Fig. 5.10 the phase diagram is shown in the (a, b)-plane, while the parameter $\Delta\omega$ was kept fixed at $\Delta\omega = 1.\text{Hz}$.

When the parameters, a, b, or $\Delta\omega$, are changed, continuous changes of the attractor states are observed. However, if a boundary line of the phase diagram is crossed, phase transitions are observed that may involve abrupt state changes, loss of stability and hysteresis. Different paths in parameter space provide for different transition routes. Several such paths are shown in Fig. 5.10 corresponding to the transition routes that have been observed in our experimental system. On path number one the state close to antiphase first becomes unstable, so that a switch to in phase occurs. The in phase pattern likewise becomes unstable and wrapping solutions are observed. Path number two corresponds to cases in which an antiphase type solution is never stable and a transition from in phase to wrapping is observed. Along path number three the in phase solution persists, but the antiphase solution becomes unstable. Path number four leads directly to wrapping solutions from a régime in which both in phase and antiphase are stable. If we also consider path number four together with path number two in reverse direction, all experimentally observed transition scenarios in the ascending condition are covered (see Results). Similarly the four paths taken in the reverse direction describe all transition scenarios in the descending condition (except for the bouts of phase wandering).

The present theoretical formulation gives rise to several predictions that can be tested in further experimental work. The most important prediction concerns the loss of stability which is at the origin of the observed transitions. This loss can be observed through stability measures such as the relaxation time of relative phase, the time it takes to relax to the pattern after a small perturbation was applied for a short period of time, or the amount of fluctuations of relative phase, measured in terms of its variability. When a transition is approached and the system is prepared in the state that loses stability (cf. Schöner, Haken, & Kelso, 1986) these measures increase strongly. One can see this intuitively in Fig. 5.9: As a state loses stability the slope of $\dot{\phi}$ at that state becomes flatter, that is, the

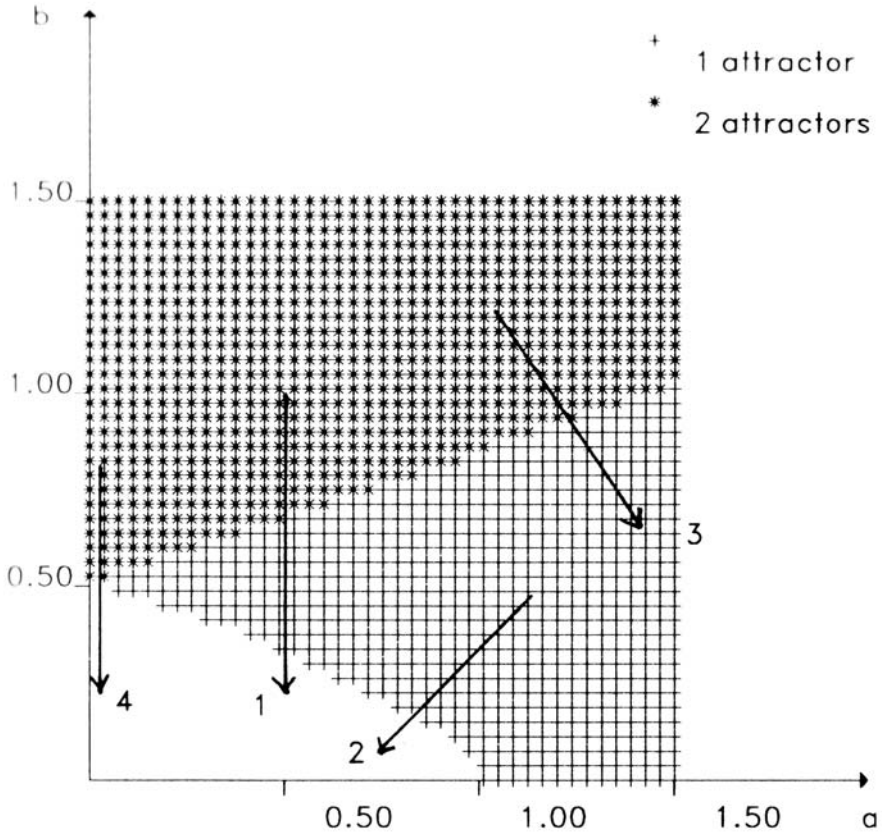


FIG. 5.10. The phase diagram of Equation (1) for $\Delta\omega = 1.$ Hz. In the different régimes the system has different phase portraits. A régime with two point attractors, one close to in phase and one close to antiphase, exists for $b >> \Delta\omega$ and $b >> a$. For $a >> \Delta\omega$ and $a >> 4b$ a monostable régime is found in which only the attractor close to in phase remains stable. For $\Delta\omega >> a$ and $\Delta\omega >> b$ no attractor exists and running solutions of Equation (1) are encountered. Different paths in the parameter plane (a, b) yield different transition routes that can be compared to the experimental results (see text). The dynamics depicted in Figure 5.9 correspond to different points on path number one.

system is "slowing down." Hints of a corresponding enhancement of fluctuations (cf. Kelso, Scholz & Schöner, 1986) are evident in the experimental data shown in Fig. 5.4. Fig. 5.11 shows the different solutions of Equation (1) as the parameter b is decreased along path number one as shown in Fig. 5.10. The relaxation time of each solution is shown and exhibits loss of stability as the transition lines are crossed (Note that for the range of

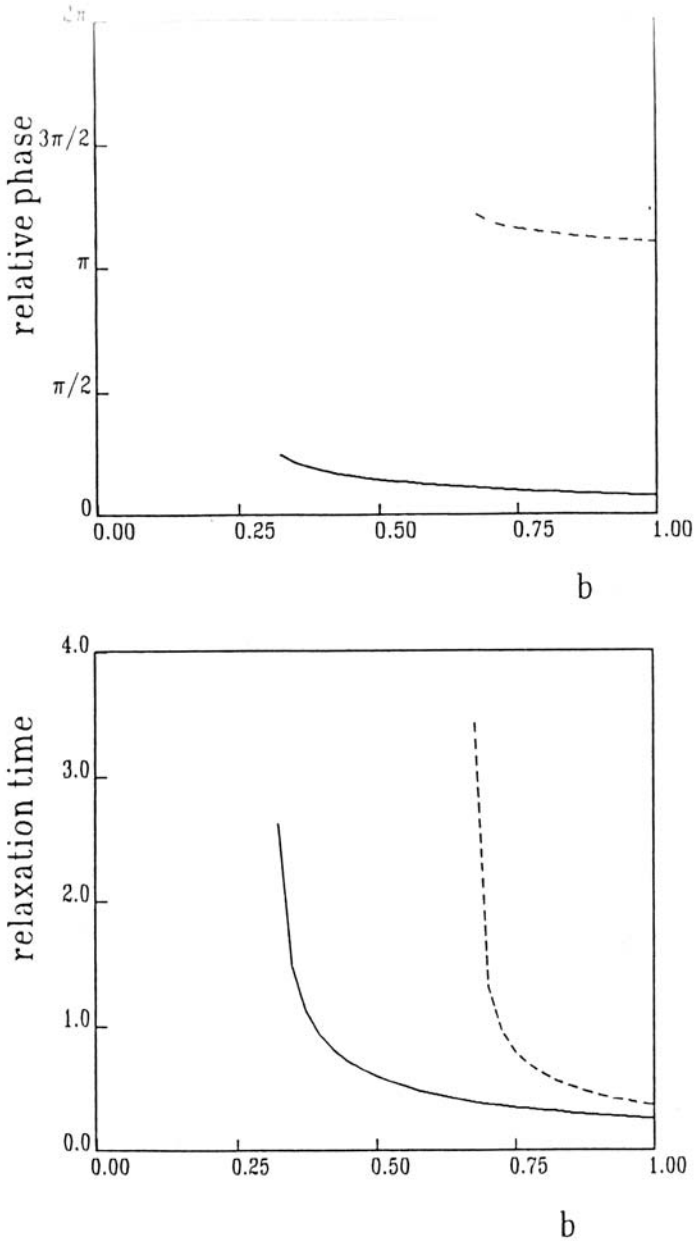


FIG. 5.11. (Top) The stationary solutions of Equation (1) and their relaxation time (bottom) as a function of the parameter b (the parameters $a = 0.5$ Hz and $\Delta\omega = 1$ Hz are kept fixed). The transition route shown corresponds to the path number one in Figure 5.10. For large b two attractors, one close to in phase ($\phi \approx 0$: dashed lines) and one close to antiphase ($\phi \approx \pi$: full line), exist (cf. left part of Figure 5.9). As b is decreased first the attractor close to antiphase, then the attractor close to in phase become unstable. The loss of stability is indicated by an increase in the relaxation times of these attractors.

the parameter plane shown, the relaxation times of the different patterns are in a realistic order of magnitude, $\tau_{rel} \approx 1$ s.). The variability of relative phase shows the same increase as can be seen from the relation (Schöner, Haken, & Kelso, 1986)

$$SD = \sqrt{Q\tau_{rel}} \tag{3}$$

where SD is the standard deviation of relative phase around its mean state. For transitions among attractors, a further measure is the length of the transient of the actual switch—the so-called switching time—that can be calculated theoretically given the stability measures (Schöner, Haken, & Kelso, 1986) and measured experimentally (Scholz, Kelso, & Schöner, 1987).

An interesting aspect emerges when the so-called running solutions are examined. In the theory, these running solutions have a fine structure: They spend more time at those relative phase values at which the force ϕ is minimal. This is the phenomenon of *relative coordination* observed first by von Holst (1937/73) in studies of rhythmically moving limbs of several species. Fig. 5.12 shows a numerical simulation of Equation (1) in this parameter régime. The observed periodic modulation of relative phase is a signature of relative coordination and is due to the attraction to certain preferred phase relationships contained in the dynamics of Equation (1). The periodicity interval of these relative phase modulations can vary considerably due to noise, illustrating that this periodicity is not, in itself, a stable state.

GENERAL DISCUSSION AND SUMMARY

Building on previous work that dealt with patterns of movement coordination, their stability and change, we studied here the question of how synchronization with a periodically structured environment occurs. We treated this problem as one of pattern formation, that is, as an attempt to understand the dynamic nature of the relationship between action and perception. A variety of different entrainment régimes between hand and metronome were observed and the pathways among patterns noted. A simple theoretical model—again developed from dynamic pattern theory—was able to reproduce the observed timing patterns. These are in turn derivable from the component level of description.

It is interesting to note that terms such as *in phase* or *antiphase* are *relative* in the system studied here. For example, as one subject (LS) so cleverly revealed, it is possible to form an *antiphase* pattern (defined experimentally as a flexion motion off-the-beat) as an *in phase* pattern, by producing an extension motion on-the-beat. The *abstract* dynamics of en-

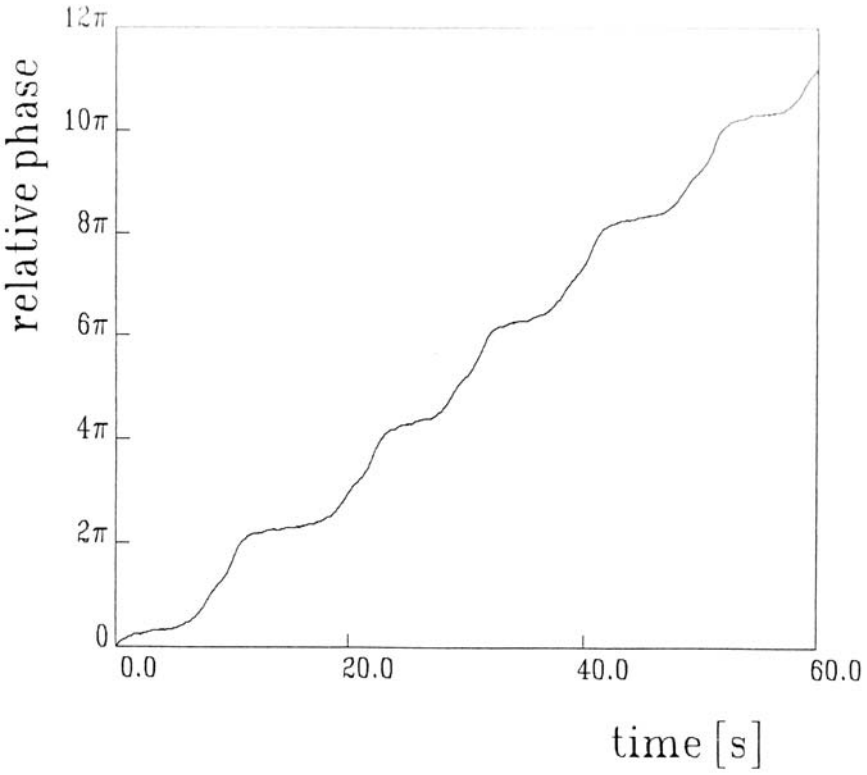


FIG. 5.12. Numerical simulation of Equation (1) at $\Delta\omega = 1.$ Hz, $a = .5$ Hz, $b = .25$ Hz, that is, in the régime in which only "running" solutions exist. Note the plateaus in the relative phase trajectory that exhibit the tendency to maintain certain preferred phase relationships.

trainment and synchronization are what matter in our action–perception paradigm: There are presumably many ways to assemble temporally ordered patterns. Nevertheless, we should emphasize that although the dynamics are abstract, they are always *measureable*, as an operational theory requires. Our modeling strategy of identifying collective variables for (in the present case) action–perception patterns, mapping them onto a minimum number of attractors and studying stability and its loss, offers a conceptual framework to understand how multiple, temporally ordered states may arise from different mechanisms. We treat the environment in a parametric fashion, that is, as that aspect of the physical environment that influences the behavioral patterns without itself being influenced by the system. For the physical environment to become meaningful to the biological system it is necessary that a dynamic relation exist between the system and its environment. We conceive this process as the formation of

an action–perception pattern. Environmental information is viewed as meaningful and specific to the extent that is forms part of the behavioral dynamics. More precisely, environmental information perturbs the vector field of the intrinsic dynamics, such that it attracts these dynamics to the required behavioral pattern (cf. Schöner & Kelso, 1988a, b, c). In the present case, the limit cycle dynamics of the hand are "tunable," in the sense that they are modified by an external periodic signal. We see also that there are limits to this process—readily interpreted as loss of entrainment. When metronome frequency is scaled, the system shifts from one entrainment region to another. In fact, our simple model reveals a variety of pathways through which different states (stable and unstable) are achieved (see Theoretical Model and discussion therein). Often, loss of stability accompanies the shift from one entrainment régime to another as evident from enhanced fluctuations of the order parameter, relative phase. Such effects are predicted from dynamic pattern theory provided certain time scales relations exist (for a readable discussion, see Kelso et al., 1987, or Schöner & Kelso, 1988d).

What are the general implications of the present theoretical strategy for understanding perceptual–motor skills? Are we dealing with a very special case here, for example? We think not. Our psychological ancestors were deeply interested in the problem we have studied here, we think with good reason. As beautifully exemplified in Lee's work (e.g., Lee & Young, 1986, for review), synchronization and relative timing are crucial aspects of skilled behavior. Unfortunately, the parametric influences of environmental information are not always explored, as in Lee and Lishman's (1975) study of body sway, where only a single visual driving frequency was employed. The possibility of identifying multiple visual–postural dynamic patterns—analogous to those found here—and mapping them onto a theoretical model (with new predictions and different observables) remains open. Lee and Lishman (1975) for example, focus on the sway velocity of the *body* as their main dependent measure, rather than, as we have done here, the relative phasing between environment and body. Their Fig. 4, which shows the time series of the moving room and the moving body embedded in that environment, could clearly be subjected to the experimental and theoretical analysis conducted here in terms of collective variables spanning the action–perception system.

"Understanding" of the action–perception system depends, of course, on the language used. Inherent in our approach is an abstraction away from *what* material objects are producing temporal patterns, and toward defining the dynamical laws governing the patterns themselves. Concepts of collective variables characterizing entrained states, control parameters, multistability, loss of stability, and so forth have only recently been demonstrated to be useful in understanding temporally patterned phenomena

at several scales of observation. A notable feature of the dynamic pattern approach is that it is possible to *derive* observed patterns and their dynamics from cooperative interactions at the component level. Our model also shows that the multiple observed patterns we have observed experimentally correspond to different régimes of the *same* underlying dynamics.

ACKNOWLEDGMENTS

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